Integrated-stress-response modulators

Scheda E01_isr_modulators · F3 ipotesi favorevole dal bersaglio, non provata in WWOX

Dati aggiornati al: 2026-10-29 · **Registro dei bisogni del** 2026-10-27 · **Catalogo costruito il** 2026-10-28T23:40:16Z (tre date diverse: l'ultimo aggiornamento della ricerca (schede, bisogni, registri dei cicli), la data del registro dei bisogni e la costruzione del catalogo dei composti).

1. Domanda (il bisogno)	Risposta integrata allo stress
2. Candidato / intervento	**Integrated-stress-response modulators (eIF2B activators of the ISRIB class, PERK-arm inhibitors, repurposed translational-repression reversers)** (Obiettivo: Beneficio parziale utile; Track A (indipendente dalla variante); ipotesi)
3. Risultato WWOX	Nessun risultato dell'intervento in un sistema WWOX; osservazione del bisogno collegato: Human neural progenitors with WWOX silenced by shRNA (CAGE transcriptome, GSE126075): the authors' gene-set report (GSEA on 9,427 genes, GO biological process, their threshold FDR q below 0.25) lists on the knockdown side 108 sets, among them amino-acid biosynthesis and metabolism (alpha-amino acid biosynthetic process q 0.065, serine family q 0.098, amino acid activation q 0.173, amino-acid transmembrane transport q 0.175), 18 rRNA, tRNA and ribosome sets (ribosome biogenesis q 0.055) and regulation of the ERAD pathway (q 0.091).
4. Precedente esterno	ISR biomarker output under an eIF2B activator: decrease – Target engagement shown for DNL343, a brain-penetrant eIF2B activator.
5. Controevidenze, per categoria	1. [sicurezza e danno] Adult ALS trial: eIF2B activation engaged its target with no clinical benefit (abstract-level); a PERK inhibitor showed signs of pancreatic toxicity in a mouse model. 2. [sicurezza e danno] The only paediatric dataset is in vanishing white matter, a disease of eIF2B itself, with serious adverse events in about three quarters of 33 children and hallucinations in over half (abstract-level); the card does not state that the agent is an eIF2B activator, and the dataset is not suitability for WWOX-DEE. 3. [controprove fuori da WWOX e limiti] ALS functional rating and survival (joint model): disease rate ratio relative to placebo: no_change – Ratio 1.04 (95 percent credible interval 0.83 to 1.32) over a 24-week placebo-controlled period: no benefit despite target engagement and proof of mechanism in the earlier phase. + 9 altre nella scheda. (ordine per categoria: prima sicurezza e danno, poi risultati nulli o sfavorevoli in sistemi WWOX, poi controprove fuori da WWOX; dentro la categoria l'ordine della scheda; non è un giudizio di peso)
6. Dato mancante	Whether any WWOX-deficient human neural system has raised eIF2 alpha phosphorylation or ATF4 protein, with or without a proteotoxic load.
7. Prossimo passo	Stage 1 (decides everything after it): eIF2 alpha phosphorylation, ATF4 protein and ATF4-target transcripts in WWOX-deficient versus isogenic human neural cells at matched maturity, with and without a defined proteotoxic challenge. If not elevated, park with the reopening condition below. Stage 2 (only if elevated): an eIF2B activator versus vehicle on that readout and on one functional endpoint. In the fly: an Atf4 loss-of-function arm with the Ldh and lactate readouts.
8. Fonti essenziali (al massimo 5)	• PMID 31543760 https://pubmed.ncbi.nlm.nih.gov/31543760/ The WWOX Gene Influences Cellular Pathways in the Neuronal Differentiation of Human Neural Progenitor Cells. [Front Cell Neurosci, 2019] • PMID 40825784 https://pubmed.ncbi.nlm.nih.gov/40825784/ Investigational eIF2B activator DNL343 modulates the integrated stress response in preclinical models of TDP-43 pathology and individuals with ALS in a randomized clinical trial. [Nat Commun, 2025] • PMID 42771346 https://pubmed.ncbi.nlm.nih.gov/42771346/ DNL343 and Disease Progression in Amyotrophic Lateral Sclerosis: A Randomized Clinical Trial. [JAMA Netw Open, 2026] • PMID 42179084 https://pubmed.ncbi.nlm.nih.gov/42179084/ Pharmacological targeting of the integrated stress response by 2BAct improves object recognition memory and reduces neuroinflammation in the 5xFAD model of Alzheimer's disease. [J Alzheimers Dis, 2026] • PMID 42586097 https://pubmed.ncbi.nlm.nih.gov/42586097/ Safety and efficacy of guanabenz in early-childhood onset vanishing white matter: primary analysis of a single-arm, phase 1/2 trial. [Lancet Neurol, 2026]

Dossier completo (3-4 pagine, appendice): #export=E01_isr_modulators. Materiale di ricerca per la discussione con il team clinico curante: non è consiglio medico e non è una graduatoria di trattamenti. Associazione non è modulazione, e modulazione non è recupero. Generato da un registro del repository (non incluso in questo pacchetto) (schema legend.wwox.repurposing_view/2, shareable). Versione condivisibile: vedi il piè di pagina della pagina per le fonti incluse.